

CCMS : Corruption Complaint Management System

Project Objective

Develop a system where citizens can:

- Register and log in
- Submit corruption complaints
- Upload evidence (images/documents)
- Track complaint status
- View complaint history

The government/admin can:

- View all complaints
- Assign officers
- Change complaint status
- Prioritize urgent complaints
- Generate reports

Project Modules

Citizen Module

- Dashboard
- File Complaint
- Complaint History
- Upload Evidence
- Track Complaint
- Notifications
- Feedback
- Profile

Officer Module

- Dashboard
- Assigned Complaints
- Investigation
- Upload Investigation Images
- Change Status
- Notes

Admin Module

- Dashboard
- Manage Users
- Manage Officers
- Manage Departments
- Manage Categories
- Manage Complaints
- Assign Complaints
- Reports
- Analytics
- Activity Logs



Citizen Panel

Dashboard

Shows

- Total Complaints
- Pending Complaints
- In Progress
- Resolved
- Rejected
- Latest Notifications

Features

Register

- Citizen Registration
- Email Verification (Optional)
- Mobile Number

Login

- Email & Password
- Forgot Password

My Profile

- Update Profile
- Change Password
- Upload Profile Photo

File New Complaint

Citizen enters

- Complaint Title
- Category
- Department
- Description
- Date
- Time
- Location

- Anonymous Complaint (Yes/No)
- Priority (optional)
- Upload Evidence

Upload

- Images
- PDF
- Video
- Audio

My Complaints

Shows

- Complaint ID
- Title
- Department
- Status
- Date

Actions

- View
- Download Receipt
- Withdraw Complaint (Before Assignment)

Track Complaint

Search by

- Complaint ID

Complaint Timeline (Example)

12 July
Complaint Submitted

↓

13 July
Assigned to Officer

↓

15 July
Evidence Verified

↓

18 July
Resolved

(Excellent place to demonstrate a Linked List.)

Evidence Gallery

View uploaded

- Images
- Documents
- Videos

Feedback

After resolution

Citizen gives

- Rating
- Comments

Notifications

Receive

- Complaint Assigned
- Investigation Started
- Status Changed
- Complaint Closed

Help & FAQ (also AI Chatbot)

Common questions

Contact Support

Support Form



Officer Panel

Officer logs in.

Dashboard

Shows

- Assigned Complaints
- Pending
- Under Investigation
- Completed
- High Priority Cases

Assigned Complaints

Table

Complaint ID
Citizen
Department
Priority
Status
Assigned Date

Complaint Details

View

- Complaint
- Evidence
- Citizen Details
- Location

Investigation

Officer can

- Start Investigation
- Add Notes
- Upload Photos
- Upload Documents

Update Status

Status

Pending



Assigned



Investigation Started



Evidence Collected



Resolved



Closed

(Each update can be stored in the **Investigation Logs** table.)

Upload Investigation Evidence

Upload

- Images
- PDF
- Audio

Investigation Notes

Example

Visited Government Office



Interviewed Staff



Verified Documents

Search Complaint

By

- Complaint ID
- Citizen Name

Priority Queue

Officer sees

Critical

↓

High

↓

Medium

↓

Low

(Perfect use of a **Priority Queue**.)

My Performance

Statistics

- Cases Assigned
- Cases Solved
- Average Resolution Time

Notifications

Receive

- New Assignment
- Deadline Reminder

Profile

Update

- Photo
- Password



Admin Panel

Main control panel.

Dashboard

Charts

- Total Complaints
 - Pending
 - In Progress
 - Resolved
 - Fake Complaints
 - Monthly Reports
-

Complaint Management

View All Complaints

Actions

- View
 - Assign
 - Delete
 - Export
-

Complaint Assignment

Assign

Complaint



Officer

This is where you can implement the **Greedy Algorithm**:

Assign the complaint to the officer with the fewest active cases (or nearest officer).

User Management

Manage

- Citizens

Actions

- Add
 - Update
 - Delete
 - Block
 - Activate
-

Officer Management

Manage

- Officers

Actions

- Add
 - Edit
 - Delete
 - Change Department
-

Department Management

Departments

Police

Revenue

Municipality

Health

Education

Transport

Complaint Categories

Manage

Bribery

Land Scam

Road Scam

Police Misconduct

Tender Fraud

Reports

Generate

- Daily Report
- Weekly Report
- Monthly Report
- Yearly Report

Export

- PDF
 - Excel
-

Analytics Dashboard

Graphs

- Complaint Trends
 - Department-wise Complaints
 - City-wise Complaints
 - Resolution Rate
 - Officer Performance
-

Search

Search

- Complaint ID
 - User
 - Officer
 - Department
-

Sorting

Sort by

- Date

- Priority
- Status
- Officer
- Department

Activity Logs

Shows

Admin Login

↓

Officer Added

↓

Complaint Assigned

↓

Department Updated

Notifications

Broadcast

- New Rules
- Public Notices

System Settings

Manage

- Website Name
- Logo
- Contact Details
- Email Settings



Permission Matrix

Feature	Citizen	Officer	Admin
Register	✓	✗	✗
Login	✓	✓	✓
File Complaint	✓	✗	✗
Upload Evidence	✓	✓	✗
View Own Complaints	✓	✗	✗
Track Complaint	✓	✓	✓
View Assigned Complaints	✗	✓	✓
Update Complaint Status	✗	✓	✓
Add Investigation Notes	✗	✓	✗
Manage Users	✗	✗	✓
Manage Officers	✗	✗	✓
Manage Departments	✗	✗	✓
Assign Complaints	✗	✗	✓
Reports & Analytics	✗	✓	✓
Activity Logs	✗	✗	✓
System Settings	✗	✗	✓



Where Each DSA Fits

DSA	FEATURE
Linked List	Complaint Timeline / Investigation History
Queue	Pending Complaint Processing
Stack	Undo Status Changes
Binary Tree	Department Hierarchy
Binary Search Tree	Complaint ID Index
Graph	Department Workflow / Complaint Routing
BFS / DFS	Department Traversal / Workflow Analysis
Linear Search	Search by Name, Category, Department
Binary Search	Search Complaint by ID
Merge Sort	Sort Complaints by Date
Quick Sort	Sort by Priority
Insertion Sort	Sort by Citizen Name
Priority Queue (Heap)	Critical Complaint Processing
Greedy Algorithm	Assign Complaint to the Least-Busy or Nearest Officer